Multimodal Image Fusion Framework for End-to-End Remote Sensing Image Registration

Liangzhi Li, Ling Han, Mingtao Ding, and Hongye Cao

***Abstract*— We formulate the registration as a function that maps the input reference and sensed images to eight displacement parameters between prescribed matching points, as opposed to the usual techniques (feature extraction–description–matching–geometric restrictions). The projection transformation matrix (PTM) is then computed in the neural network and used to warp the sensed image, uniting all matching tasks under one framework. In this article, we offer a multimodal image fusion network with self-attention to merge the feature representation of the reference and sensed images. The integration information is then utilized to regress the prescribed points' displacement parameters to get PTM between the reference and sensed images. Finally, PTM is supplied into the spatial transformation network (STN), which warps the sensed image to the same coordinates as the reference image, achieving end-to-end matching. In addition, a dual-supervised loss function is proposed to optimize the network from both the prescribed point displacement and the overall pixel matching perspectives. The effectiveness of our method is validated by qualitative and quantitative experimental results on multimodal remote sensing image matching tasks. The code is available at: https://github.com/liliangzhi110/E2EIR.**

***Index Terms*— End-to-end registration, multimodal fusion, remote sensing image, spatial transformer networks.**

I. Introduction

THE complementarity of multimodal images can lead to a thorough comprehension of the observed scene, and the fusion or joint analysis of these two image types can greatly improve remote sensing's interpretative ability. Thus, there is a primary task of multimodal image matching, which is required for remote sensing image spatial analysis. Even though automatic image registration methods have made huge progress in recent decades, multimodal image matching remains a difficulty due to nonlinear radiometric differences [1], [2], [3].

Area- and feature-based image registration methods are two types of remote sensing image registration algorithms. Area-based methods start by extracting a collection of candidate feature points from the image, then searching for their correspondence within the search window for each feature point. The relationship between local intensities, i.e., sum of squared differences (SSD) [4], normalized cross-correlation (NCC) [5], and mutual information (MI) [6], is used to measure the similarity. However, they are sensitive to imaging conditions, image distortion, and speckle noise, and have particularly high computational complexity, limiting their use for multimodal image matching.

For feature-based methods, they extract salient features of the image, including point, line, and surface features, and measure their feature description used to obtain correspondences. Feature descriptions mainly use local structural information, where the keypoints with the smallest distances are the best matches. Various feature-based methods have been developed over the past decades, represented by the scale-invariant feature transform (SIFT) [7], including these improved SIFT methods, such as SURF [8], PCA-SIFT [9], synthetic aperture radar (SAR)-SIFT [10], and DSP-SIFT [11]. However, a plethora of studies have demonstrated that the aforementioned techniques are inadequate in effectively addressing the complexities of multimodal image matching [1]. Furthermore, the utilization of nonlearning-based feature matching techniques in the context of multimodal image registration is inadequate in effectively addressing the complex and highly nonlinear correspondence issues that arise from the diversity of imaging settings and the resulting variations in nonlinear metrics.

Currently, there is a trend toward the utilization of data-driven deep learning (DL)-based methods for advanced image feature extraction in the context of multimodal image matching. Wang et al. [12] developed a collection of DL frameworks that are based on point feature descriptions for the task of remote sensing image registration. Ye et al. [13] tried to extract fusion features by using convolutional neural networks (CNNs) and SIFT. Ma et al. [14] used the depth feature extracted from CNN and the local feature to improve the proportion of correct correspondence. Li et al. [2] proposed a novel interaction fusion module for the purpose of generating feature representations through the fusion of information derived from two distinct images. Zhang et al. [15] proposed a self-supervised optical flow fine-tuning network framework to densely register in optical images and SAR images.

Meanwhile, many methods use DL to map the similarity of the patch matching. For example, Hughes et al. [16] proposed a network framework for synthetic aperture radar (SAR) and optical image matching, predicting the region of each image that is suitable for matching. Fang et al. [17] developed a DL model for SAR-optical image matching based on a twinned

Manuscript received 18 May 2022; revised 8 September 2022, 15 January 2023, and 4 February 2023; accepted 19 February 2023. Date of publication 22 February 2023; date of current version 25 April 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 42171348. *(Corresponding author: Ling Han.)*

Liangzhi Li, Mingtao Ding, and Hongye Cao are with the College of Geological Engineering and Geomatics, Chang'an University, Xi'an, Shaanxi 710054, China.

Ling Han is with the School of Land Engineering, Chang'an University, Xi'an 710064, China (e-mail: hanling@chd.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3247642

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

U-Net with a fast Fourier transform (FFT) correlation layer. Hughes et al. [18] used a pseudo-twin CNN to identify the corresponding SAR-optical patches. Li et al., [19] proposed a semantic template matching framework that is capable of outputting centroid position matching. Xiang et al. [20] proposed a feature decoupling network based on template features for description and matching in both optical and SAR images.

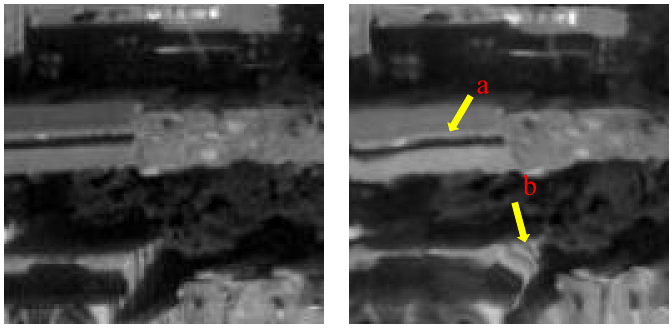
Although these methods are effective in many cases, it is time-consuming caused by the tedious detection–description–matching steps. In addition, these methods need to be considered for applicability to different sensor images. To this end, we present a multimodal image fusion framework for end-to-end remote sensing image registration, avoiding feature detection-description-geometry constraint processing. First, we develop a remote sensing image fusion network for integrating the cross-modal feature representation between multimodal images by using the self-attention mechanism. Then, the fully connected layer is used to fuse the two branch features for regressing the prescribed point displacement parameters. Subsequently, a transformation matrix between two images is calculated internally. Finally, the spatial transformation network (STN) [21] is used to warp the sensed image to the same coordinates as the reference image, achieving end-to-end matching. Furthermore, we propose a dual-supervised loss function for optimizing the network from both the prescribed point displacement and the overall pixel matching perspectives. The main contributions are summarized as follows.

1) We define registration as a function that maps the input image pair to the displacement of a specified point, which we use in conjunction with the STN network to distort the sensed image to accomplish end-to-end remote sensing image matching.
2) A multimodal remote sensing image fusion network is designed to integrate the feature representation between reference and remote sensing images by using a densely connected network (DenseNet) [22] and an attention mechanism, which is used to regress the prescribed point displacement parameters between the sensed image and the reference image.
3) We propose a dual-supervised loss function with mean square error (mse) and normalized MI (NMI), in which mse is used to optimize the prescribed point displacement loss and NMI is applied for quantifying all pixel matching.

The rest of this article is arranged as follows. The second part describes the work related to end-to-end registration and attention mechanisms. III presents our methods and datasets. The registration performance of our method and other methods for remote sensing images is presented in Section IV, followed by a summary in Section V.

II. Related Work

This section first introduces studies related to end-to-end matching of images, followed by a review of the attention mechanism.



(a) (b)

Fig. 1. (a) Reference image. (b) Sensed image after registration obtained by regressing the deformation field method. The yellow arrow represents the wrong matching area.

A. End-to-End Registration

End-to-end registration methods are first proposed for natural and medical images, such as [23], [24], [25], [26], and [27]. Those end-to-end image registration methods are based on image similarity metrics to train its network. More specifically, the reference and sensed images are mapped image matching by DL networks to regress the deformation field or projection transformation matrix (PTM), and then combined with STN to distort the sensed image to obtain end-to-end matching. However, these methods based on deformation fields and PTMs are challenging to apply to complicated remote sensing images, because each pixel in remote sensing images generally contains precise geospatial location data and the grayscale information of remote sensing images with multi-phase is nonlinear.

Fig. 1 shows the matching drawbacks of the end-to-end registration method based on regression of deformation fields (proposed by [28]) on multi-temporal remote sensing images. The roofs of the two structures are substantially deformed, as illustrated in Fig. 1(a) and (b), and inconsistent similarity assessments resulted in improper transformations, resulting in incorrect matching. This approach is challenging to complex and variable multimodal remote sensing images because it establishes displacement connections for each pixel. For end-to-end registration methods based on the PTM, they fail to optimize both rotation and translation. For example, T_θ is as follows:

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \\ \theta_{31} & \theta_{32} & \theta_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \tag{1}$$

$$x' = \frac{u}{w}, \quad y' = \frac{v}{w} \tag{2}$$

where $\theta_{11}, \theta_{12}, \ldots, \theta_{33}$ constitute the PTM.

It should be noted that the submatrix $[\,\theta_{11}\ \theta_{12}\ \theta_{21}\ \theta_{22}\,]$ represents the rotational terms in the PTM, while the vector $[\,\theta_{13}\ \theta_{23}\,]$ represents the translational offset. Balancing and synchronously optimizing these two terms is a challenging task for the network. Therefore, we introduce the displacement of the prescribed point as the regression parameter, and the detailed procedure will be described in the next Section III.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

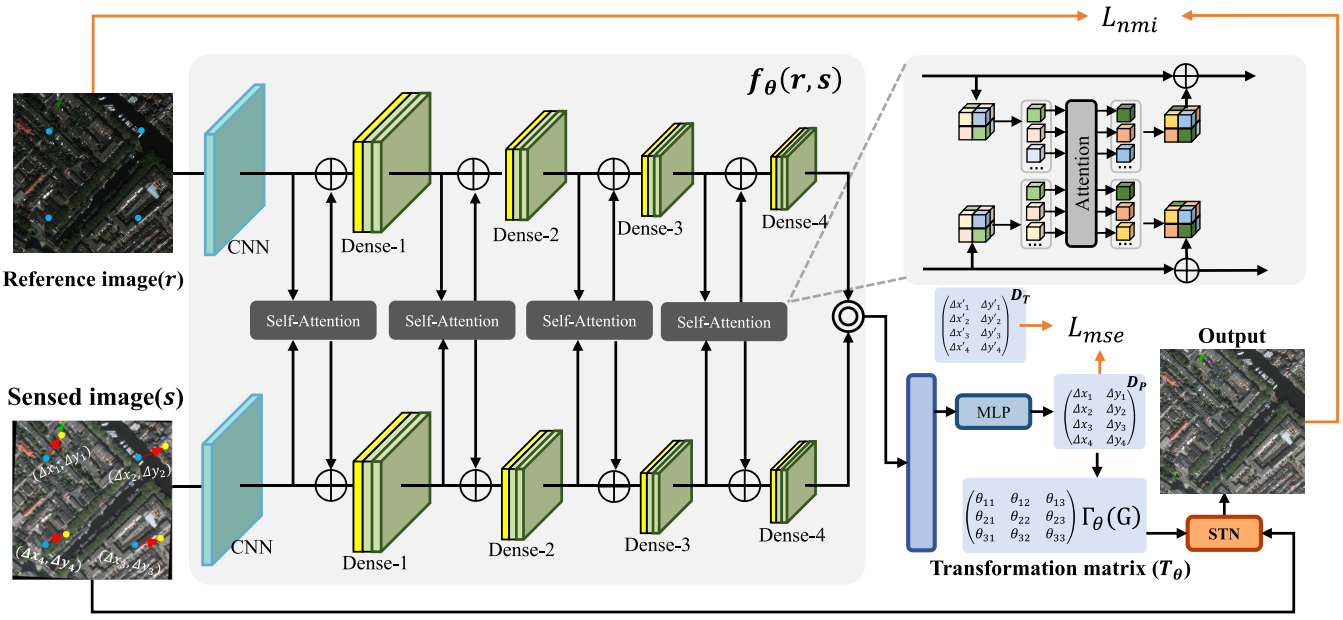


Fig. 2. Overview of the proposed method. The $f_\theta(r, s)$ network models the displacement parameters(D_p) between fixed and moving points, where the blue and yellow points indicate fixed and moving points, respectively. Projection transformation parameters (T_θ) are calculated by D_p at the end of the network. STN combined with T_θ is used to warp the sensed image to obtain the output image(o). L_{mse} is applied to optimize the loss of the ground true(D_T) and displacement parameters(D_p). L_{nmi} is applied for quantifying all pixel matching. $\oplus$ denotes the feature summation.

B. Attention Mechanism

The nonlinear radiometric variations that exist between multimodal remote sensing images pose significant challenges for the use of networks with identical weights or pseudo-twins for cross-modal feature extraction, as it becomes difficult to obtain feature similarity that can effectively express the displacement relationships between the images. In addition, the limited field of perception of CNNs fails to capture global information. This results in the inability to obtain long-range positional dependencies between multimodal remote sensing images. Recently, Visual Transformer (ViT) used by self-attention outperforms many CNN structures in vision tasks due to its ability to capture global information. Compared to CNN, ViT relies on excellent modeling capabilities to achieve very good performance on several benchmarks such as ImageNet [29], COCO [30], and ADE20k [31].

Based on a self-attention mechanism, the proposed method uses internal data and feature correlation to gain a bigger perceptual domain and contextual information by recording global data. To increase the information interaction between the reference and sensed image, we adopt a self-attention interactive fusion mechanism. For mapping the displacement relationship between the two images, information fusion between the two channels is used to generate a feature representation with positional pixel fusion. When compared to convolutional operations, extracting local features frequently employs the method of stacking convolutional layers to expand the perceptual area, which adds to the network fitting cost. The self-attention mechanism uses a triplet of (key, query, value) to model global contextual information, with key and query being points propagated to get the corresponding attention weights and the obtained weights being point propagated with value to get the final output. The detailed process is described in Section III.

III. Proposed Method

A. Overview of the Method

We specify the pixel points at fixed positions in each reference image as matching points, and the role of the network model is to regress the matching point displacements ($[\Delta x_1, \Delta y_1, \Delta x_2, \Delta y_2, \Delta x_3, \Delta y_3, \Delta x_4, \Delta y_4]$, as shown in Fig. 2). In our experiments, the fixed positions we used are $(20, 20)$, $(236, 20)$, $(236, 236)$, and $(20, 236)$. We will describe the choice of fixed points in detail in this section.

An overview of the proposed method is presented in Fig. 2. The input consists of a reference image r and a sensed image s. We utilize a dual channel DenseNet combined with self-attention $[f_\theta(r, s)]$ for regressing the displacement (D_p) of the matching point. The projection transformation parameters (T_θ) between r and s are calculated by the displacement parameters inside the network. STN combined with T_θ are used to warp the sensed image to $s \cdot T_\theta$ (output image), allowing us to assess the alignment degree of $s \cdot T_\theta$ and r. Two loss functions ($L_{\text{mse}}, L_{\text{nmi}}$) are proposed. The first captures mse loss between the ground true value and the predicted 8 displacement parameters, while the second captures all pixel alignment degrees between r and o images.

B. Network Architecture

The overall network structure is composed of parallel dense blocks and vertical self-attention structures. The parallel structure is composed of a convolutional layer and four densely connected blocks. The output of each densely connected block

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

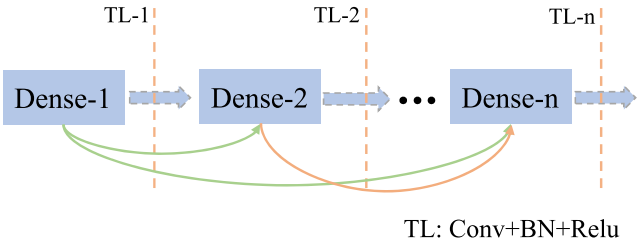


Fig. 3. Densely connected networks, the output of all the previous layers is input to the later layer, which enables the network to send the gradient to the network more quickly with feedback. Any two adjacent layers are connected directly by multiple operations such as BN, rectified linear unit (ReLU), and convolution layer, where BN + Conv + ReLU denotes the transformation layer.

is interactively fused using the self-attention mechanism. The fully connected layer is used to integrate the parallel network features for regressing the eight displacement parameters in four layers with the number of neurons 1024, 512, 128, and 8. The following is a full description of the densely connected block and self-attention.

1) Densely Connected Network: The architecture of dense blocks is shown in Fig. 3, where any two consecutive layers are directly connected through a series of operations, including batch normalization (BN) [32], ReLU [33], and convolution layers. The notation BN + Conv + ReLU denotes a transformation layer. The output of all the previous layers is input to the later layer, which enables the network to send the gradient to the network more quickly with feedback. The output of layer l is defined as

$$x_l = H_l\left(\left[x_{l-1}, x_{l-2}, x_{l-3}, \ldots, x_0\right]\right) \tag{3}$$

where H_l denotes a module composed of BN, ReLU, convolution, and dropout layers. The notation $[x_{l-1}, x_{l-2}, x_{l-3}, \ldots, x_0]$ represents the concatenation of feature maps output from layers 0 to $l-1$, and x_l is the output feature map of layer l.

When each layer of the network produces k feature maps as output, the total number of input feature maps for the i layer is $k \times (i-1) + k_0$, where k_0 is the number of channels of the input. While each layer only produces k output feature maps, the total number of feature maps after reuse for each dense block is still relatively large. To reduce the number of feature maps in the transition layer, it is necessary to employ feature compression after each dense block. The specific parameter settings for this process are outlined in Table I.

2) Self-Attention: A self-attention mechanism is employed for fusing feature information between the reference and sensed images by considering the global context. A succession of discrete tokens, each represented by a feature vector, is sent into the self-attention module. A position code is added to this feature vector to account for the position sensing bias. Before each dense block, we combine the information from the two channels, where the series of discrete tokens consists of patches with sizes of 8×8. As shown in Fig. 4, the structure of a self-attention layer involves the linear transformation of input vectors $a^1, a^2, a^3 \in \mathbb{R}^{d_l \times 1}$ using weight

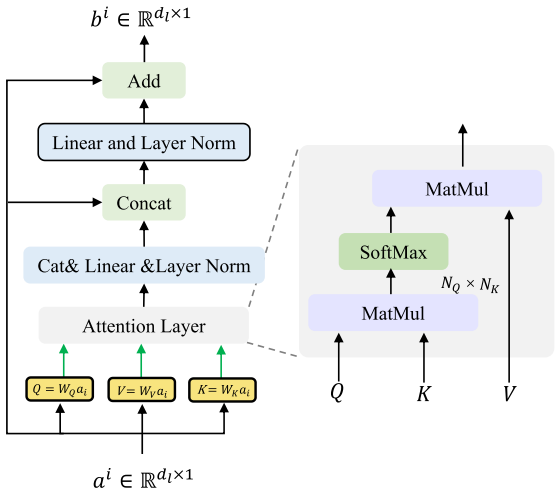


Fig. 4. Attention layer in proposed multimodel fusion framework.

TABLE I

ARCHITECTURE OF BACKBONE. THE ABBREVIATIONS ARE GIVEN AS FOLLOWS. CONV: CONVOLUTION LAYER. DENSE: DENSE BLOCK. TRANS: TRANSITION LAYER

Layer	Structure parameters	Operation ($k = 8$)
Conv-0	256×256×64	3×3 conv
Dense-1	256×256×72	(1×1 conv; 3×3 conv) × 6
Trans-1	128×128×64	1×1 conv; 2×2 average pool
Dense-2	128×128×80	(1×1 conv; 3×3 conv) × 12
Trans-2	64×64×64	1×1 conv; 2×2 average pool
Dense-3	64×64×96	(1×1 conv; 3×3 conv) × 28
Trans-3	32×32×64	1×1 conv; 2×2 average pool
Dense-4	32×32×128	(1×1 conv; 3×3 conv) × 28
Trans-4	16×16×128	1×1 conv; 2×2 average pool

matrices $W^q \in \mathbb{R}^{d_k \times d_l}$, $W^k \in \mathbb{R}^{d_k \times d_l}$, $W^v \in \mathbb{R}^{d_l \times d_l}$ to obtain Q ($q^i \in \mathbb{R}^{d_k \times 1}$), K ($k^i \in \mathbb{R}^{d_k \times 1}$), and V ($v^i \in \mathbb{R}^{d_l \times 1}$), where $i \in 1, 2, 3$

$$\begin{cases} q^i = W^q \cdot a^i \\ k^i = W^k \cdot a^i \quad i \in \{1, 2, 3\} \\ v^i = W^v \cdot a^i. \end{cases} \tag{4}$$

Let $A = (a^1, a^2, a^3) \in \mathbb{R}^{d_l \times 3}$, $Q = (q^1, q^2, q^3) \in \mathbb{R}^{d_k \times 3}$, $K = (k^1, k^2, k^3) \in \mathbb{R}^{d_k \times 3}$, $V = (v^1, v^2, v^3) \in \mathbb{R}^{d_l \times 3}$, then Q, K, V

$$\begin{cases} lQ = W^q \cdot A \\ K = W^k \cdot A \\ V = W^v \cdot A. \end{cases} \tag{5}$$

Then Attention(Q, K, V) is calculated using the obtained Q, K, V [34]. The dot product scaling is formulated as follows.

$$\alpha_l^i = \frac{\left(q^i\right)^\top \cdot k^l}{\sqrt{d^k}} = \frac{\sqrt{d^k}}{d^k} \sum_{n=1}^{d^k} k_n^l \cdot q_n^i, \quad i, l \in \{1, 2, 3\}. \tag{6}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

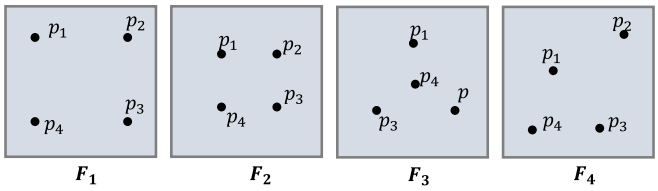


Fig. 5. Illustration of prescribed fixed matching points for regressing the displacement parameters. F_1, F_2, F_3, and F_4 denote four different prescribed fixed points, where p_1, p_2, p_3, and p_4 are the positions of the fixed points, respectively.

Assume that $k^l = (k_1^l, k_2^l, k_3^l)$ and $q^i = (q_1^i, q_2^i, q_3^i)$ are independently distribution and their mean and variance are 0 and 1.

The mean value of the l component α_l^i of the attention vector $a^i \in \mathbb{R}^{3\times 1}$ is 0 and the variance is 1. The formula is as follows:

$$\mathbb{E}\left[\alpha_l^i\right] = \frac{\sqrt{d^k}}{d^k}\sum_{n=1}^{d^k}\mathbb{E}\left[k_n^l\right]\cdot\mathbb{E}\left[q_n^i\right] = 0, \quad i,l \in \{1,2,3\}$$
$$\mathrm{Var}\left[\alpha_l^i\right] = \frac{1}{d^k}\sum_{n=1}^{d^k}\mathrm{Var}\left[k_n^l\right]\cdot\mathrm{Var}\left[q_n^i\right] = 1, \quad i,l \in \{1,2,3\}. \tag{7}$$

Let $\Lambda = (\alpha^1, \alpha^2, \alpha^3) \in \mathbb{R}^{3\times 3}$ be the attention fraction matrix, then Λ is

$$\Lambda = \frac{K^\top \cdot Q}{\sqrt{d^k}}. \tag{8}$$

The attention fraction vector α^i is passed through the $SoftMax$ layer to obtain β^i, which is

$$\beta_j^i = \frac{e^{\alpha_j^i}}{\sum_{n=1}^3 e^{\alpha_n^i}}, \quad i,j = \{1,2,3\}. \tag{9}$$

Finally, using β^i and V obtain the final output $b^i \in \mathbb{R}^{dl\times 1}$, b^i is

$$b^i = \sum_{l=1}^{3}\beta_l^i \cdot v^l, \quad i \in \{1,2,3\}. \tag{10}$$

Let the output matrix $B = (b^1, b^2, b^3) \in \mathbb{R}^{d_l\times 3}$, and then B

$$B = \mathrm{Attention}(Q,K,V) = V\cdot\mathrm{softmax}\left(\frac{K^\top\cdot Q}{\sqrt{d^k}}\right). \tag{11}$$

The attentional features obtained above are then used to obtain the final output b^i according to layers in Fig. 4.

C. Prescribed Point Displacement

We use the network to regress the displacements for prescribed points, where the fixed points can be random but are discrete. As shown in Fig. 5, there are four options to specify the points F_1, F_2, F_3, F_4. In Section IV, we will exploit the effect of F_1, F_2, F_3, F_4 on matching accuracy. The PTM between two images can be calculated by matching point displacement. The formula is given by

$$D_p = \left[\Delta x_1, \Delta y_1, \Delta x_2, \Delta y_2, \Delta x_3, \Delta y_3, \Delta x_4, \Delta y_4\right] \tag{12}$$

where Δx_1 and Δy_1, denote the displacement of a single point in the sensed image on the x-axis and y-axis, respectively, relative to the reference image

$$x' = x + \Delta x, \quad y' = y + \Delta y \tag{13}$$

where x and y represent the pixel coordinates in the sensed image and x' and y' represent the pixel coordinates of the image after registration.

It should be emphasized that any four fixed nonlinear points can be selected as matching points according to the actual needs, and the exact position of the matching points is not our focus. The transformation matrix is calculated in the form of (15), (16) and (17):

$$t = \begin{pmatrix} 0 & 0 & 0 & -x & -y & -1 & xy' & yy' & y' \\ x & y & 1 & 0 & 0 & 0 & -xx' & -yx' & -x' \\ -xy' & -yy' & -y' & xx' & yx' & x' & 0 & 0 & 0 \end{pmatrix} \tag{14}$$

$$\theta = (\theta_{11}\ \theta_{12}\ \theta_{13}\ \theta_{21}\ \theta_{22}\ \theta_{23}\ \theta_{31}\ \theta_{32}\ \theta_{33})^{\mathrm{T}} \tag{15}$$

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = t\cdot\theta \tag{16}$$

where $\theta_{11}, \theta_{12}, \ldots, \theta_{33}$ is the PTM.

D. Spatial Transformer Networks

The STN is used to learn the translation, rotation, and scaling invariance of feature maps. It can be used not only to change intermediate feature maps but also to interpolate output images. To obtain the final alignment of the image, we employ the STN to warp sensed images to produce the transformation of pixel coordinates from the sensed image to the reference image, as shown in Fig. 6.

Localization networks, a grid generator, and a sampler make up STN. For parameter regression prediction (localization networks), they are implemented by the proposed network. The grid generator realizes pixel coordinate correspondence, and we extend the degree of freedom of image transformation from affine to projection transformation, as defined by (17), (18) and (19). Pixel interpolation conversion is implemented by the sampler. To achieve the differentiability of the coordinate solution in the experiment, we apply bilinear interpolation. The formula's details are as follows:

$$V_i^c = \sum_n^H\sum_m^W U_{nm}^c \max(0, 1-|x_i - m|)\max(0, 1-|y_i - n|) \tag{17}$$

where (x_i, y_i) coordinates represent the spatial position of the input image, U_{nm}^c represents the value in color channel c with coordinates (n, m), and and V_i^c is the output value for pixel i at location (x_i', y_i') in channel c. To allow the backpropagation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

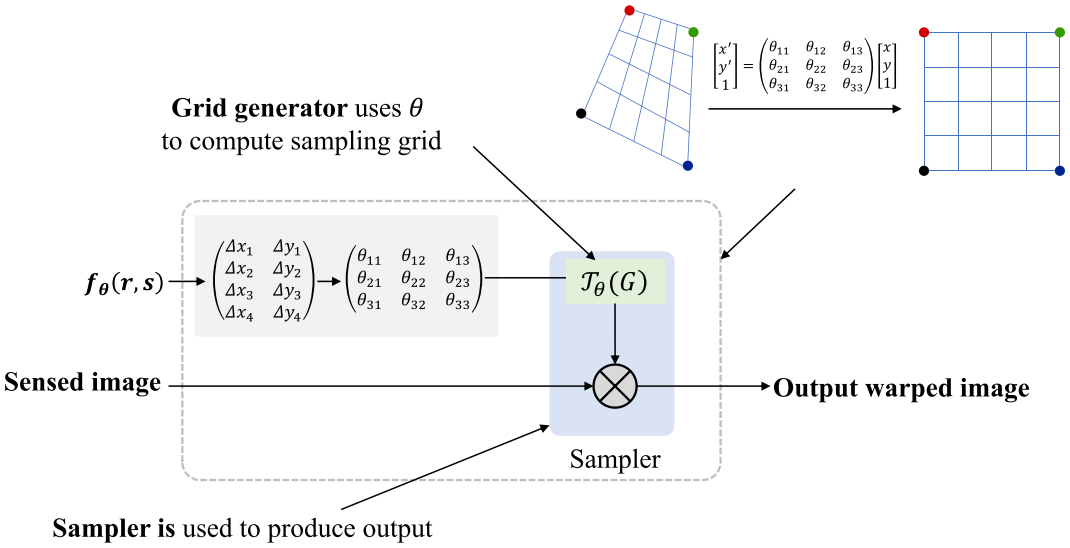


Fig. 6. Example of applying the STN to sensed images producing the output. where θ is the transformation parameters.

loss, we can calculate the partial derivative of the function:

$$\frac{\partial V_i^c}{\partial U_{nm}^c} = \sum_n^H \sum_m^W \max(0, 1 - |x_i - m|) \max(0, 1 - |y_i - n|) \tag{18}$$

$$\frac{\partial V_i^c}{\partial x_i} = \sum_n^H \sum_m^W U_{nm}^c \max(0, 1 - |y_i - n|) = \begin{cases} 0, & \text{if } |m - x_i| \geq 1 \\ 1, & \text{if } m \geq x_i \\ -1, & \text{if } m < x_i. \end{cases} \tag{19}$$

E. Loss Function

The final loss consists of the prescribed point displacement loss (L_{mse}) and overall pixel matching loss (L_{nmi}).

L_{mse} evaluates the difference between the predicted matching point displacement and the ground truth. L_{mse} is the Euclidean distance as defined in [35]. The loss function is given by

$$L_{\text{mse}} = \frac{1}{N} \sum_{i=1}^{N} (\text{Net} - D_4)^2 \tag{20}$$

where N is the number of training samples and Net indicates the predicted value.

The degree of alignment between reference and warped sensed images is captured using L_{nmi}. MI refers to the amount of information contained in one random variable about another random variable, and it is a common way to figure out how similar two patches are. Therefore, the L_{nmi} loss function is proposed for determining the degree of alignment for image matches

$$L_{\text{nmi}} = \frac{\sum_a P_A(a) \log P_A(a) + \sum_a P_B(b) \log P_B(b)}{\sum_{a,b} P_{AB}(a, b) \log P_{AB}(a, b)}. \tag{21}$$

In (21), A represents the reference image and B represents the sensed image. The joint distribution of the intensities of the two images is represented by $p(a, b)$, and the marginal probability distributions of gray are represented by $p(a)$ and $p(b)$. As $p(a)p(b)$ represents the distribution of pixel intensities if the two images are independent, the least similarity between $p(a, b)$ and $p(a)p(b)$ is observed when the two images A and B are aligned. This is because this configuration maximizes the amount of information one image provides about the other.

The final loss function is

$$\text{Loss} = \alpha \cdot L_{\text{mse}} + \beta \cdot L_{\text{nmi}}, \quad \alpha + \beta = 1 \tag{22}$$

where α and β are the two coefficients satisfying $\alpha \geq 0$, $\beta \geq 0$ and $\alpha + \beta = 1.0$. In the initial training stage, the sample data are used to train the network, and taking a larger α value is used to accelerate the network convergence and obtain the displacement parameters of coarse image matching ($\alpha = 0.8$ and $\beta = 0.2$). Then, in the later training stage, we use a large number of samples to fine-tune the registration network to refine the multi-modal image matching with different sensors ($\alpha = 0.2$ and $\beta = 0.8$).

F. Dataset and Training

For training, validating, and testing the proposed network, we employ the PS-RGB (RGB), Multispectral (MS), and SAR-intensity (SAR) datasets from the SpaceNet [36] dataset. These data contain different scenes such as ports, commercial complexes, and suburbs. Capella Space provided the SAR images via an aerial sensor, while WorldView-2 provided the optical images. For SAR images, their data in the x-band have four polarizations (HH, HV, VH, and VV), where the VV polarization is used for training and testing. We stitch the images in the dataset first and then uniform interpolate to $1m$ resolution because there are blank places in the images. We use the above three data for matching to obtain three datasets RGB $\leftrightarrow$ MS, RGB$\leftrightarrow$SAR, and MS$\leftrightarrow$SAR.

Since true image pair displacement is difficult to get by hand, we develop a sample pair generating method that can obtain true matching point displacement. As seen in Fig. 7,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

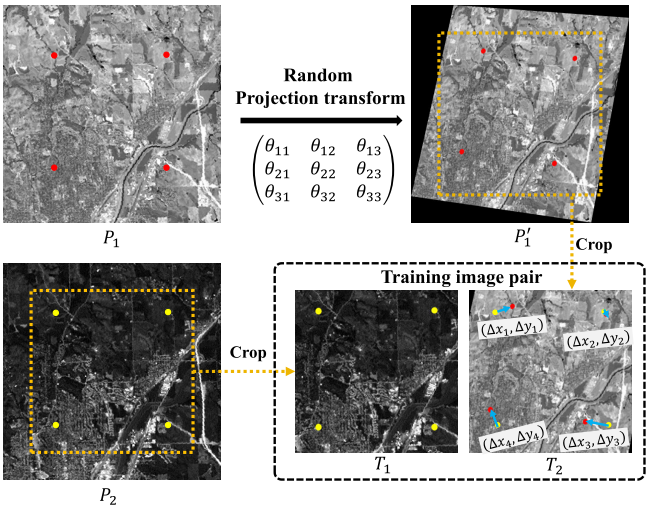


Fig. 7. Overview of generating training images, where P_1 and P_2 are two precisely aligned images. We use the randomly generated projection transformation (θ) equation to warp P_1 to obtain P_1' and then crop the same positions of P_1' and P_2 to obtain T_1 and T_2 as the training sample image pair. The displacement of the matching points $(\Delta x_1, \Delta y_1)$, $(\Delta x_2, \Delta y_2)$, $(\Delta x_3, \Delta y_3)$, $(\Delta x_4, \Delta y_4)$ are used as the ground-truth values.

the specific generation process is described. The PTM is first generated randomly within a specific range, and then used to distort one of the pixel-aligned images P_1 and P_2. As a registration pair, the matching image patch is cropped between the reference and sensed image. Finally, the true regression sample value $(\Delta x, \Delta y)$ is used to determine the matching point displacement. To improve the model's robustness, we create a training dataset that comprises both positive (positive samples mean sensed images contain four matching points in the reference image) and negative samples (negative samples indicate that the sensed images do not contain four matching points in the reference image and may contain 0, 1, 2 or 3).

From SpaceNet, datasets with a total number of 450 000 are generated. The experiments in this article are performed on an AMAX workstation with Ubuntu 18.04 LTS, an RTX3090Ti GPU, and 128-GB RAM, with an initial learning rate of 0.0001 and an epoch number of 320.

IV. Experimental Result

In this section, we first introduce the evaluation metrics used to measure matching accuracy, compare the effect of four prescribed point displacements (F_1, F_2, F_3, and F_4) on matching accuracy, present an ablation study to highlight the importance of the different components of our model, and finally, compare the performance of the proposed method with several baselines.

A. Evaluation Protocol

We use metrics including root-mean-square error (RMSE) and MI to evaluate image registration in our experiments. The registration accuracy of four matching points and two match patches is measured using RMSE and MI, respectively.

1) Root-Mean-Square Error: The RMSE is used to evaluate the matching accuracy between the matching point displacement and the predicted value. The formula is given as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\left(x_i' - x_i''\right)^2 + \left(y_i' - y_i''\right)^2\right)} \tag{23}$$

where n is 4, which means four matching point pairs, and (x_i'', y_i'') denotes the transformed coordinates of (x_i', y_i').

2) Mutual Information: MI considers an image's gray level to be a spatially uniform stochastic process with independent samples, with statistical characteristics and probability density functions describing the image's statistical qualities. To evaluate the similarity measurement between two image pairs, MI is employed for examining the statistical dependence of two images. MI is defined as follows for two images X and Y:

$$\mathrm{MI}(X, Y) = H(X) + H(Y) - H(X, Y) \tag{24}$$

where $H(X)$ and $H(Y)$ represent the information entropy of the two images and $H(X, Y)$ represents the joint entropy of the two images

$$H(X) = -\sum_{i=0}^{255} p_X(i)\log(p_X(i)) \tag{25}$$

$$H(X, Y) = -\sum_{i=0}^{255}\sum_{j=0}^{255} p_{XY}(i, j)\log(p_{XY}(i, j)) \tag{26}$$

where $p_X(i)$ represents the probability distribution of grayscale for X and $p_{XY}(i, j)$ represents the joint probability distribution of grayscale for X and Y.

B. Effect of the Specified Point Position

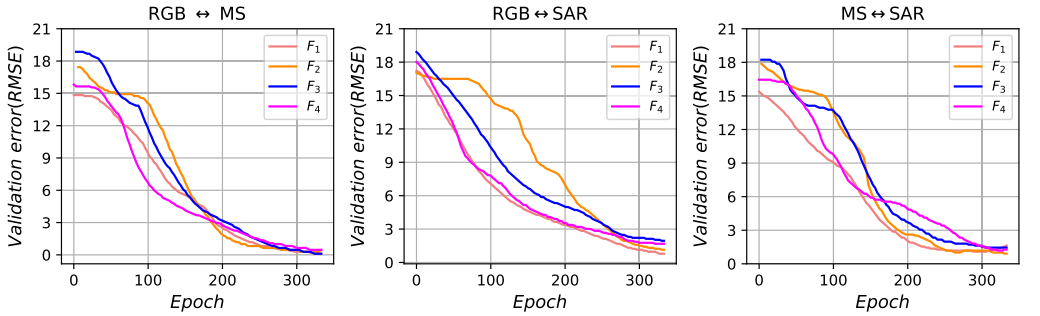
Although the prescribed point's location is determined at random, it is vital to choose the point as far as possible. This is because the displacements of the four nearby spots on the perceptual image may be modest, making learning the exact transformation relationships challenging for the network. On RGB ↔ MS, RGB↔SAR, MS↔SAR, we compare four discretely specified points(F_1, F_2, F_3, F_4).

To ensure a fair comparison, the hyperparameters for model training are kept consistent, including batch size, epoch time, and learning rate. RMSE is used to evaluate the matching performance of the validation data. Fig. 8 shows the validation accuracy on RGB ↔ MS, RGB↔SAR, MS↔SAR. The validation loss of F_1, F_2, F_3, and F_4 has the same trend on the three datasets, and the RMSE gradually decreases as the epoch increases, which achieves better matching accuracy at the end. The result demonstrates that the four specified points' placements can be random if they are guaranteed to be discrete. The following trials are all carried out by F_1.

C. Ablation Study

To help design the network described in Section III, we conduct an ablation study to compare the performance of joining various network architectures and dual-supervised loss

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.



Type	D_1	D_2	D_3
F_1	1.625	2.731	2.813
F_2	1.643	2.816	2.716
F_3	1.439	2.741	2.936
F_4	1.537	2.901	2.826

Fig. 8. Evaluation on RGB ↔ MS, RGB↔SAR, MS↔SAR image pairs for F_1, F_2, F_3, and F_4. The RMSE of the validation data is used to evaluate the performance of the matches. We also report the average matching accuracy of F_1, F_2, and F_3, and F_4 on RGB ↔ MS, RGB↔SAR, MS↔SAR. The results show that there is little difference in precision between RGB ↔ MS, RGB↔SAR, and MS↔SAR, and all of them can obtain good matching results.

TABLE II

NETWORK CONFIGURATION USED IN THE ABLATION STUDY. THE USE OF A SPECIFIC NETWORK, DENSENET, SELF-ATTENTION, L_{MSE} OR L_{NMI} IS INDICATED BY A YES (Y) OR NO (N)

Network	DenseNet	self-attention	L_{MSE}	L_{nmi}
$CL_{MSE}(N_1)$	N	N	Y	N
$DL_{MSE}(N_2)$	Y	N	Y	N
$DSL_{MSE}(N_3)$	Y	Y	Y	N
$DSL_{nmi}(N_4)$	Y	Y	N	Y
$DSL_{MSE}L_{nmi}(N_5)$	Y	Y	Y	Y

TABLE III

INFLUENCE OF DENSENET, SELF-ATTENTION, L_{MSE}, AND L_{NMI} ON THE MATCHING PERFORMANCE OF THE CORRESPONDENCE NETWORK

	RGB ↔ MS		RGB↔SAR		MS↔SAR	
	RMSE	MI	RMSE	MI	RMSE	MI
N_1	1.366	1.216	2.796	0.301	2.916	0.196
N_2	1.316	1.221	2.647	0.331	2.807	0.243
N_3	1.207	1.218	2.561	0.324	2.659	0.261
N_4	2.979	1.129	3.149	0.171	3.641	0.116
N_5	1.144	1.467	2.431	0.366	2.522	0.292

functions. STN is central to obtain remote sensing image matches. Thus, we chose STN, multilayer CNNs, as the basis of the network. We perform different combinations between DenseNet, self-attention, L_{mse}, and L_{nmi}. The results of the configuration combination are presented in Table II. The training process is designed such that the random parameters, such as epoch, learning rate, and batch size, are deterministic. The performance of the proposed method is evaluated using RMSE and MI between the matching locations. Table II provides a summary of the findings.

From Table III, the results illustrate that the average matching accuracy of N_3 is higher than that of N_2 because self-attention increases the information fusion between multimodal remote sensing images, allowing them to see each other's feature expressions. Furthermore, combining both L_{mse} and L_{nmi} loss functions further improves the matching accuracy, implying that there is a mutual feedback relationship between our proposed dual supervision and the matching point displacement loss and that reducing the matching point displacement loss can reduce the L_{nmi} loss. Therefore, N_5 is

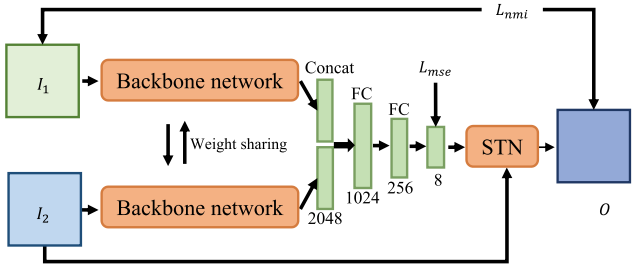


Fig. 9. Flowchart of backbones network training. The reference and sensed images are input to the weight-sharing backbone network separately. After fusing the network features of the two channels, multiple fully connected networks are followed to extract features. Finally, the two loss functions are combined to optimize the network and regress the matching point displacement parameters.

chosen as our matching framework, and all further experiments are performed on this result.

Following that, we train the dataset and test the effectiveness of our proposed overall architecture using current state-of-the-art backbones. Our proposed network architecture integrates sensed and reference images from two channels by using attention. However, current two-channel network frameworks are applied for special tasks. To use the best backbones, we make these in a generic comparative way. The total architecture of using these backbones is shown in Fig. 9. The sensed and reference images (I_1 and I_2) are sent into a backbone-based two-channel network, which is subsequently mapped to the fully connected layer using a weight-sharing strategy. Simultaneously, the fused features are regressed to the displacements of the four points on the x, y-axis on the fully connected layer. Finally, STN is used to distort the sensed image for auto-alignment, utilizing the L_{mse}, L_{nmi} loss function to optimize the network parameters.

As can be seen in Table IV, the matching accuracy of these networks is lower than that of our method, especially on RGB↔SAR, MS↔SAR. These networks are fit to labels from our entire dataset. Although they can fit our dataset well, they can only be applied for matching on RGB ↔ MS with strong texture. In contrast, the proposed method considers the cross-modal fusion between images from the feature extraction stage and establishes a mapping relationship in which the two-channel features depend on each other. This illustrates the effectiveness of the proposed multimodal fusion network.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
MATCHING RESULTS ON RGB ↔ MS, RGB↔SAR, AND MS↔SAR USING CURRENT STATE-OF-THE-ART BACKBONES

	RGB ↔ MS		RGB ↔SAR		MS ↔SAR	
	RMSE	MI	RMSE	MI	RMSE	MI
VGG-16 [37]	1.395	0.982	2.492	0.210	2.939	0.222
GoogLeNet [38]	1.479	0.873	2.511	0.179	2.887	0.223
ResNet-50 [39]	1.326	0.814	2.569	0.245	3.359	0.128
Inceptionv3 [40]	1.309	0.934	2.849	0.189	3.199	0.176
Resnetv2 [41]	1.365	0.884	2.823	0.338	2.801	0.174
ResNeXt [42]	1.693	0.815	2.881	0.186	3.411	0.101
Darknet53 [43]	1.767	0.800	2.893	0.175	3.173	0.134
CSPNet [44]	1.514	0.826	2.882	0.189	3.485	0.106
VoVNet [45]	1.394	0.957	2.527	0.223	3.110	0.231
SENet [46]	1.405	0.868	2.608	0.190	2.891	0.121
EfficientNet [47]	1.508	0.839	3.074	0.171	3.353	0.119

D. Effectiveness of L_{mse}, L_{nmi}

In the x- and y-axis direction, the L_{mse} loss function optimizes the average displacement loss of the four matching points. L_{nmi} uses the entire image's pixel alignment and feeds the L_{nmi} loss into the network structure to improve the matching point displacement parameters. Fig. 10 shows the complementarity of L_{mse} and L_{nmi} functions, with Fig. 10(a) indicating that one of the matching points is abnormally far away, while the other three matching points are pixel aligned, and Fig. 10(b) shows the matching points' displacements are all within a short range. However, L_{mse} in Fig. 10(a) and (b) are equal, with $L_{\text{nmi}} > L'_{\text{nmi}}$. This implies that L_{nmi} is sufficient to detect anomalous displacement of matching points from global image pixels, and that points with bigger displacement from the network global are penalized. Individual phenomena are avoided in the aggregate, resulting in a modest matching error at each matching point. In the test experiments, the aforesaid conclusions are also confirmed.

In the experiments, we assign different weight combinations $C_1 = (0, 1)$, $C_2 = (0.2, 0.8)$, $C_3 = (0.4, 0.6)$, $C_4 = (0.6, 0.4)$, $C_5 = (0.8, 0.2)$, and $C_6 = (1.0, 0)$ for L_{mse} and L_{nmi}, respectively, to evaluate the optimization ability on the network. These combinations are all trained by using the same hyperparameters. To evaluate the overall performance of l_{nmi}, we perform statistical analysis on 72,000 × 8 displacement parameters, and calculate the mean and variance, respectively. The detailed experimental results are shown in Table V, where the bold indicates the optimal results.

From Table V, it can be seen that on the C_6 combination, the mean is smaller and the variance is larger. Conversely, on the C_5 combination, the variance is the smallest and does not differ much from the smallest mean. It indicates that the matching performance is best under the C_5 combination and the L_{nmi} function is globally optimized. The obtained value is unusually large with the C_1 combination, and it may be difficult to make the network fit using only l_{nmi} without l_{mse} loss, function.

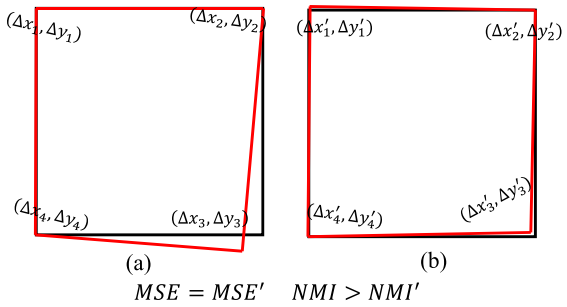


Fig. 10. L_{nmi} loss function complements the L_{mse} in the network training process. (a) and (b) has the same L_{mse} loss values, while L_{nmi} is different. L_{nmi} optimizes the network from overall pixel alignment to avoid anomalies.

TABLE V
RESULTS OF THE OVERALL AVERAGE RMSE AND MI VALUES OF 16 REGISTRATION PAIRS ON THE SPACENET DATASET

	C_1	C_2	C_3	C_4	C_5	C_6
Average value	11.436	2.953	2.867	2.864	2.862	2.853
Standard deviation	4.336	0.091	0.085	0.087	0.067	0.086

E. Experimental Results

To evaluate the performance of the proposed method for multimodal remote sensing image matching, an experimental study was conducted using 48 pairs of sensed and reference images from the test dataset. These images were selected to encompass a wide range of nonlinear radiometric variations. The RMSE and MI of 48 pairs of image experiments on the test data are shown in Fig. 11 and 12. To evaluate the matching accuracy, we compare the proposed method with baselines (SIFT, SAR-SIFT [10], CFOG [48], RIFT [49], D2-Net [50], and LoFTR). For SIFT and SAR-SIFT, the Euclidean distance ratio between the nearest and second nearest neighbors of the corresponding features is used, and RMSE is chosen for comparison. For RIFT and CFOG, we set the parameters and algorithms given in [48] and [49]. For D2-Net, LoFTR, we use their trained and shared models[1],[2] on SpaceNet for fine-tuning.

1) Quantitative Comparisons: Fig. 11 shows the results of RMSE. It can be seen that the proposed method achieves the best overall performance on 48 pairs of images on RGB ↔ MS, RGB↔SAR, and MS↔SAR. For SIFT, SAR-SIFT, CFOG, RIFT, D2-Net, and LoFTR, they obtain better accuracy on RGB ↔ MS. Although their methods are robust on RGB ↔ MS, the drawbacks of their methods as cross-modal matching can be seen on RGB↔SAR, MS↔SAR with nonlinear radiation differences. For SIFT and SAR-SIFT, their matching results have similar trends, which suggests that these methods apply to specific image matching. Predictably, our method is overall smoother than the CFOG and RIFT implementation. The latter uses detectors capable of obtaining cross-modal matches, and this approach inherently requires tuning the parameters used, which limits the matching of multi-modal images. The matching accuracy of D2-Net is lower than our method. We suspect that RGB↔SAR and

[1]https://github.com/mihaidusmanu/d2-net
[2]https://github.com/zju3dv/LoFTR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

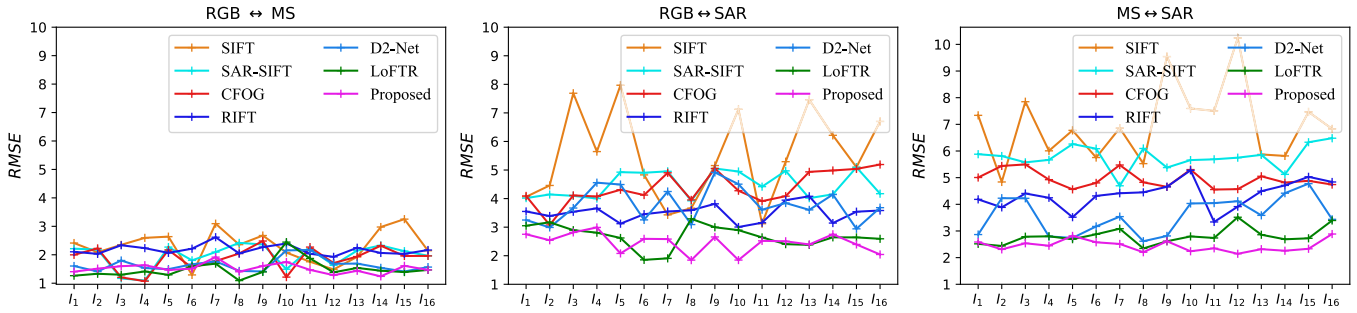


Fig. 11. Results of RMSE values of 16 registration pairs on RGB ↔ MS, RGB↔SAR, and MS↔SAR.

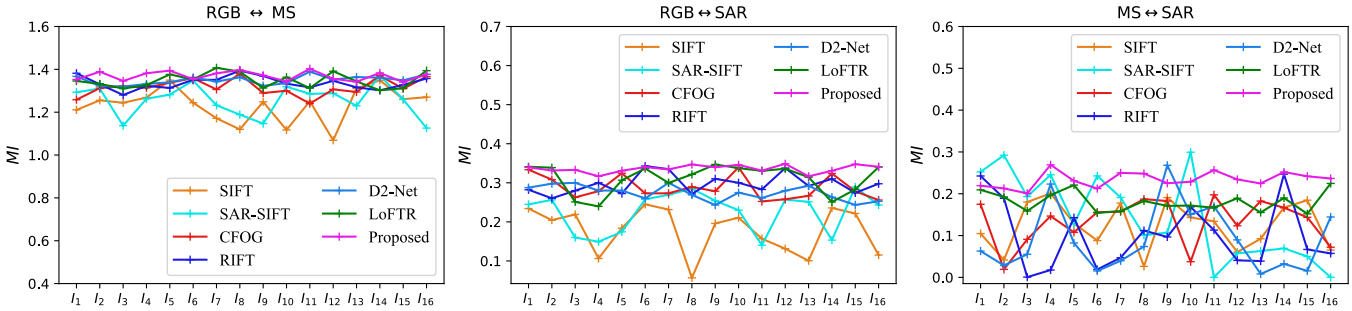


Fig. 12. Results of MI values of 16 registration pairs on RGB ↔ MS, RGB↔SAR, MS↔SAR.

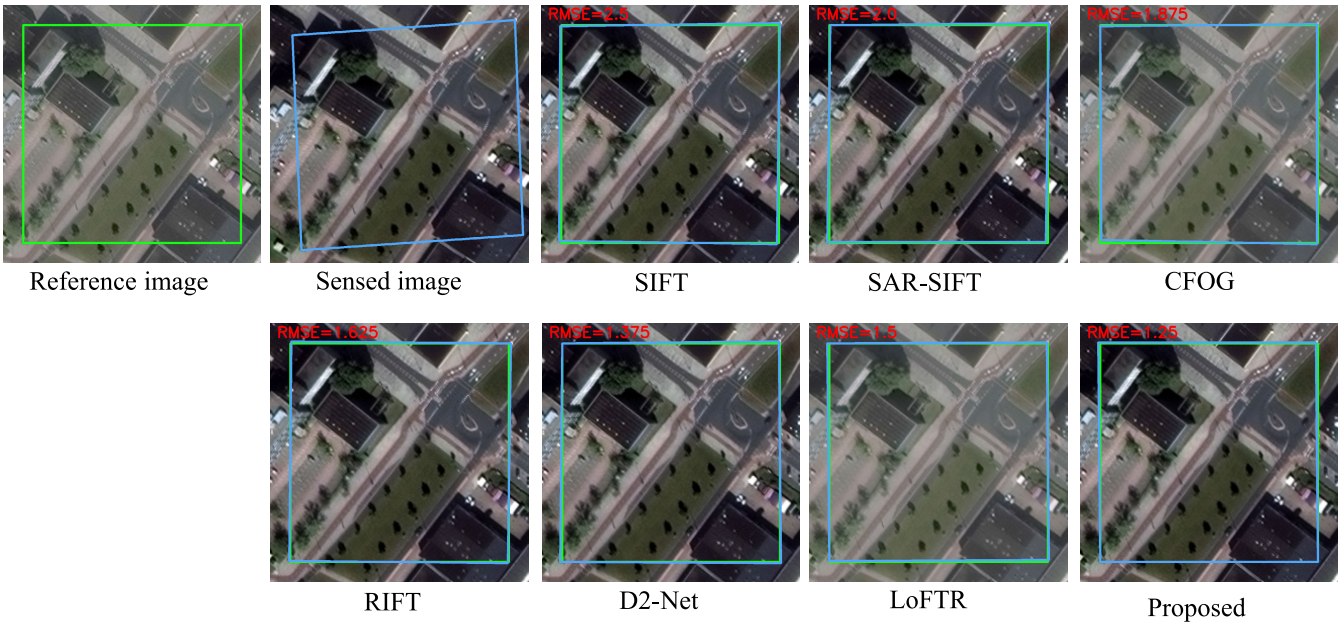


Fig. 13. Examples of matching visualizations on RGB ↔ MS. The green box indicates the reference position and the blue box indicates the matching result, where the better the overlap between the two boxes the higher the matching accuracy.

MS↔SAR matching performance with radiometric differences is poorer in D2-Net because more significant feature points cannot be obtained on these images. For LoFTR, it can obtain more matching correspondence by Transformer [34] to obtain the information interaction between the reference and sensed image. However, the random object changes and radiation differences between multimodal remote sensing images make LoFTR have more error correspondences ($\geq$ 4pixel) on the test dataset.

The MI of the test image matches is shown separately in Fig. 12. Although the trends of RMSE and MI are similar, when focused on the findings of individual registration pairs, there is little disagreement between MI and RMSE. Land cover changes may have an impact on the MI measure. Overall, our proposed method outperforms the examined methods in terms of information correlation, proving the efficiency of our proposed method in image matching.

Overall, the proposed method achieves average RMSE values of 1.144, 2.431, and 2.722 on the test datasets, which is higher than the comparison method in terms of overall matching accuracy. We select one matching pair from each of the three datasets as a visual display of the matching.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

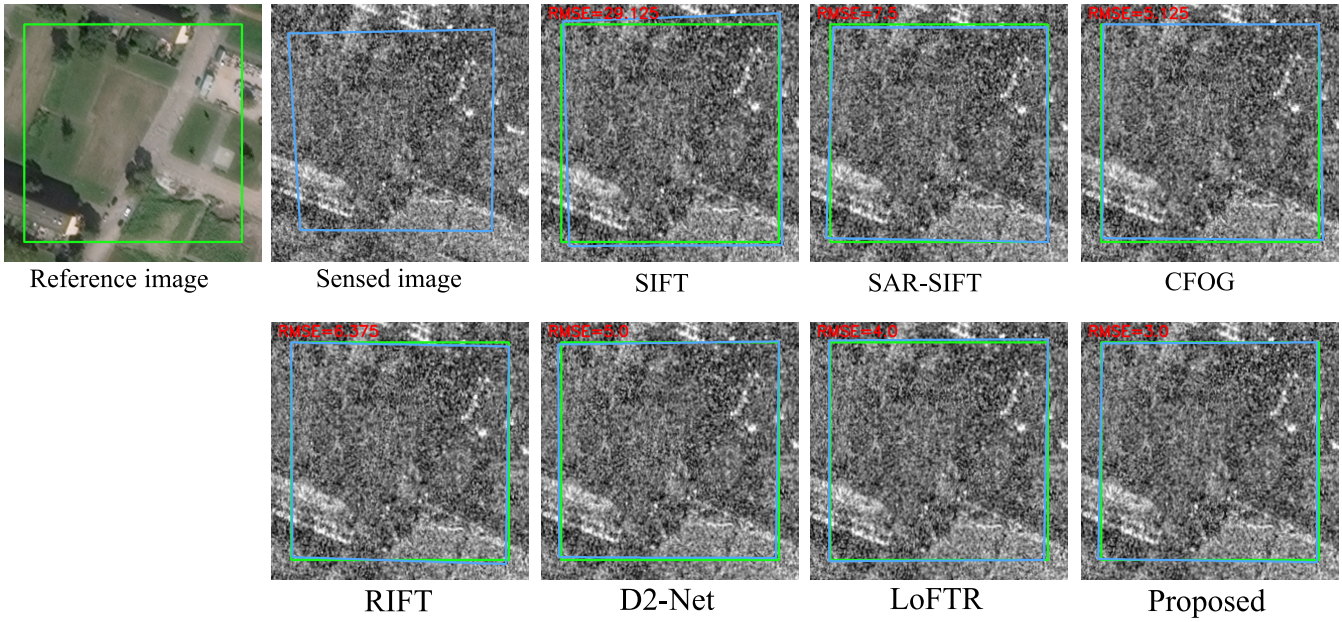


Fig. 14. Examples of matching visualizations on RGB ↔SAR. The green box indicates the reference position and the blue box indicates the matching result, where the better the overlap between the two boxes the higher the matching accuracy.

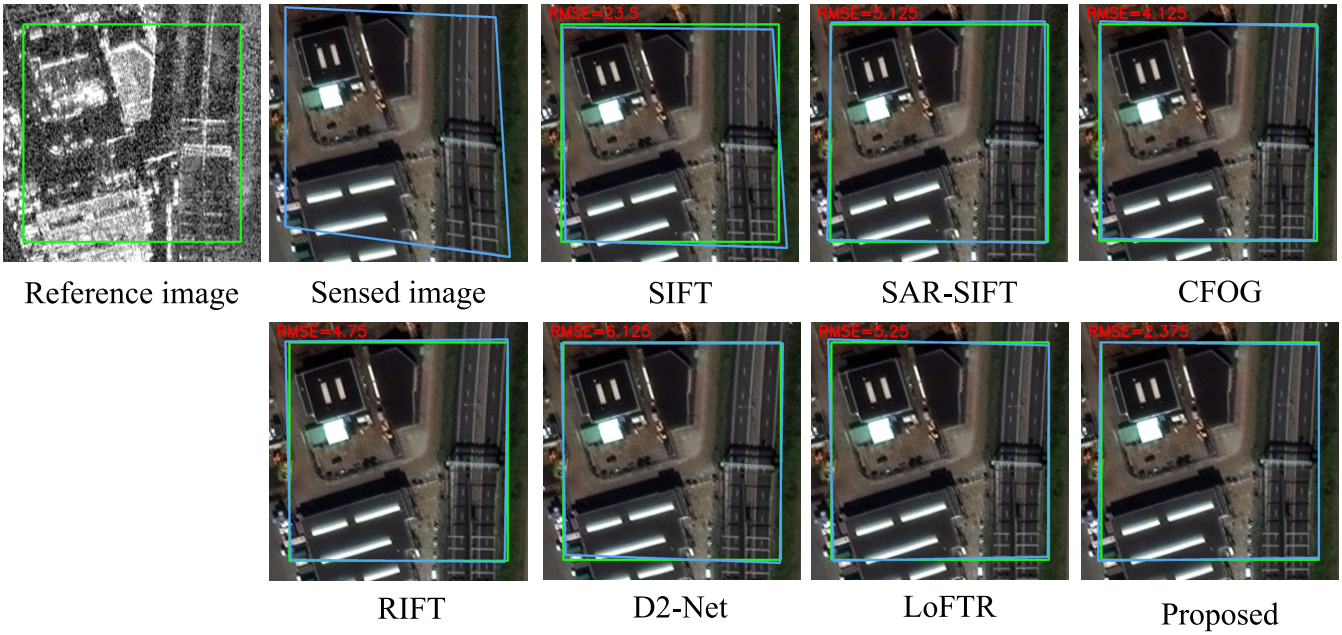


Fig. 15. Examples of matching visualizations on MS ↔SAR. The green box indicates the reference position and the blue box indicates the matching result, where the better the overlap between the two boxes the higher the matching accuracy.

Figs. 13, 14, and 15 show the matching results of the three image pairs, where the green and blue boxes represent the reference position and matching results, respectively. The registration results of SIFT and SURF methods are poor due to the nonlinear radiometric differences of multimodal remote sensing images, where the green and blue boxes are heavily crossed. The green and blue boxes of our proposed method show a large overlap in the three images, which proves the effectiveness of multimodal remote sensing image registration.

Finally, we evaluate the complexity of the models. All evaluations are based on an AMAX workstation with Ubuntu 18.04 LTS, an RTX3090Ti GPU, and 128-GB RAM. We use the entire test dataset to evaluate the average matching speed (sec/sample) for each method. On the GPU, we compare the complexity of the model using floating-point operations (FLOPs) and multiply–accumulate (MAC) operations. From Table VI, the results show that the deep neural network-based methods all have faster matching speeds on the CPU than the nonlearning SIFT, SAR-SIFT, CFOG, and RIFT. However, all these learning-based matching methods have competitive matching performance when augmented by the GPU. The reduction of matching time is contingent upon the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

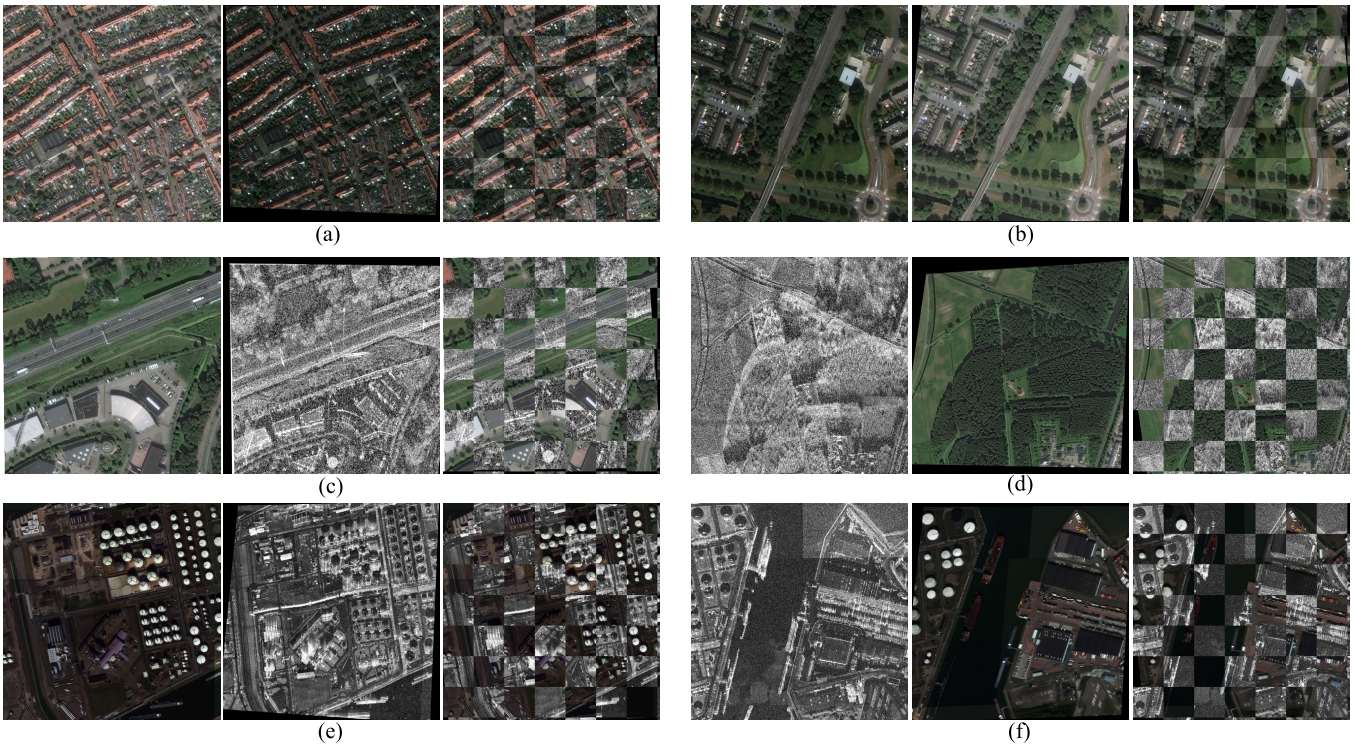


Fig. 16. Qualitative comparison results of the proposed method. (a) RGB↔MS. (b) MS↔RGB. (c) RGB↔SAR. (d) SAR↔RGB. (e) MS↔SAR. (f) SAR↔MS.

TABLE VI
TIME COMPLEXITY COMPARISONS ON THE SPACENET DATASET

Methods	SIFT	SAR-SIFT	CFOG	RIFT	D2-Net	LoFTR	Proposed	D2-Net	LoFTR	Proposed
Backend	CPU							GPU		
Average (sec.)	2.78	5.25	4.46	2.85	6.72	6.16	6.96	1.26	1.76	1.82
StandDev (sec.)	0.49	0.64	0.10	1.09	1.50	0.43	0.12	0.14	0.59	0.19
MACs	-	-	-	-	-	-	-	26.61 M	173.89 M	209.43 M
FLOPs	-	-	-	-	-	-	-	41.15 G	136.93 G	147.81 G

computational capabilities of the GPU. The results reveal that both FLOPs and MACs are higher in D2-Net and LoFTR in comparison to the proposed method. In addition, the matching speed of the proposed method is lower than that of D2-Net and LoFTR, which may be attributed to the deeper layers utilized in the proposed method.

2) Qualitative Comparison: We selected six typical images for qualitative evaluation, RGB ↔ MS, MS ↔ RGB, RGB↔SAR, SAR↔RGB, MS↔SAR, and SAR↔ MS, which provide intuitive experimental results for the algorithm. The matching results of our proposed method and the other five comparative methods are shown in Fig. 16(a)–(f). We display these matched images superimposed and select a distinct feature with a characterized matching result as an evaluation of the degree of alignment of the match. In comparison to existing approaches, our proposed method has two advantages. To begin, the proposed method considers the entire image's pixels for matching. Although the buildings on the selected area are not aligned, as shown in Fig. 16(a), (c), and (f), the proposed method can consider overall pixel matching at other locations. The lack of alignment in this location could be attributed to geometric distortion issues generated by the sensor shot's angle. Our matched images align better in Fig. 16(b), (d), and (e).

The quantitative and qualitative evaluations above demonstrate the effectiveness of the proposed method for remote sensing image registration, particularly in multimodel images.

V. CONCLUSION

In this article, we propose a fusion network for multimodal remote sensing image matching. In this network, a fusion framework based on a self-attention mechanism is used to obtain cross-modal feature representation between the sensed and reference images. The network defines the registration as a function for regressing the prescribed point displacement and warps the sensed image within the network, providing for end-to-end matching. In addition, we propose a dual-supervised loss function, and the experimental results show that it is interdependent from the optimization of individual point matching to total pixel alignment. The qualitative and quantitative results validate the effectiveness and universality of the proposed method on RGB ↔ MS, RGB↔SAR, and MS↔SAR datasets. The experimental results show that the proposed method achieves better matching results, especially

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

for RGB↔SAR and MS↔SAR corresponding to cross-modal matching.

Given that the proposed method is limited to optimal matching on fixed image sizes, which may limit the application of the method to images with larger scenes, it would be interesting to further explore the network with the input of large-scale images to achieve end-to-end remote sensing image matching.

ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers for their helpful comments and good suggestions.

REFERENCES

[1] X. Jiang, J. Ma, G. Xiao, Z. Shao, and X. Guo, "A review of multimodal image matching: Methods and applications," *Inf. Fusion*, vol. 73, pp. 22–71, Jan. 2021.

[2] L. Li, M. Liu, L. Ma, and L. Han, "Cross-modal feature description for remote sensing image matching," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102964.

[3] L. Li, L. Han, M. Ding, Z. Liu, and H. Cao, "Remote sensing image registration based on deep learning regression model," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2020.

[4] M. B. Hisham, S. N. Yaakob, R. A. A. Raof, A. B. A. Nazren, and N. M. Wafi, "Template matching using sum of squared difference and normalized cross correlation," in *Proc. IEEE Student Conf. Res. Develop. (SCOReD)*, Dec. 2015, pp. 100–104.

[5] W. Burger and M. J. Burge, *Principles of Digital Image Processing: Core Algorithms*. Cham, Switzerland: Springer, 2010.

[6] J. Walters-Williams and Y. Li, "Estimation of mutual information: A survey," in *Proc. Int. Conf. Rough Sets Knowl. Technol.* Cham, Switzerland: Springer, 2009, pp. 389–396.

[7] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, 2004.

[8] H. Bay, T. Tuytelaars, and L. V. Gool, "SURF: Speeded up robust features," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2006, pp. 404–417.

[9] Y. Ke and R. Sukthankar, "PCA-SIFT: A more distinctive representation for local image descriptors," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2004, p. 2.

[10] F. Dellinger, J. Delon, Y. Gousseau, J. Michel, and F. Tupin, "SAR-SIFT: A SIFT-like algorithm for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 453–466, Jan. 2015.

[11] J. Dong and S. Soatto, "Domain-size pooling in local descriptors: DSP-SIFT," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 5097–5106.

[12] S. Wang, D. Quan, X. Liang, M. Ning, Y. Guo, and L. Jiao, "A deep learning framework for remote sensing image registration," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 148–164, Nov. 2018. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0924271617303891

[13] F. Ye, Y. Su, H. Xiao, X. Zhao, and W. Min, "Remote sensing image registration using convolutional neural network features," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 232–236, Feb. 2018.

[14] W. Ma, J. Zhang, Y. Wu, L. Jiao, H. Zhu, and W. Zhao, "A novel two-step registration method for remote sensing images based on deep and local features," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4834–4843, Jul. 2019. [Online]. Available: https://ieeexplore.ieee.org/document/8648479/

[15] H. Zhang et al., "Optical and SAR image dense registration using a robust deep optical flow framework," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1269–1294, 2023.

[16] L. H. Hughes, D. Marcos, S. Lobry, D. Tuia, and M. Schmitt, "A deep learning framework for matching of SAR and optical imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 166–179, Nov. 2020.

[17] Y. Fang, J. Hu, C. Du, Z. Liu, and L. Zhang, "SAR-optical image matching by integrating Siamese U-Net with FFT correlation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.

[18] L. H. Hughes, M. Schmitt, L. Mou, Y. Wang, and X. X. Zhu, "Identifying corresponding patches in SAR and optical images with a pseudo-siamese CNN," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 5, pp. 784–788, May 2018.

[19] L. Li, L. Han, M. Ding, H. Cao, and H. Hu, "A deep learning semantic template matching framework for remote sensing image registration," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 205–217, Nov. 2021.

[20] D. Xiang, Y. Xie, J. Cheng, Y. Xu, H. Zhang, and Y. Zheng, "Optical and SAR image registration based on feature decoupling network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5235913.

[21] M. Jaderberg et al., "Spatial transformer networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 1–9.

[22] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2261–2269.

[23] S. Ghosal and N. Ray, "Deep deformable registration: Enhancing accuracy by fully convolutional neural net," *Pattern Recognit. Lett.*, vol. 94, pp. 81–86, Jul. 2017.

[24] G. Wu, M. Kim, Q. Wang, Y. Gao, S. Liao, and D. Shen, "Unsupervised deep feature learning for deformable registration of mr brain images," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2013, pp. 649–656.

[25] G. Balakrishnan, A. Zhao, M. R. Sabuncu, A. V. Dalca, and J. Guttag, "An unsupervised learning model for deformable medical image registration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9252–9260.

[26] A. V. Dalca, G. Balakrishnan, J. Guttag, and M. R. Sabuncu, "Unsupervised learning for fast probabilistic diffeomorphic registration," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2018, pp. 729–738.

[27] C. Stergios et al., "Linear and deformable image registration with 3D convolutional neural networks," in *Image Analysis for Moving Organ, Breast, and Thoracic Images*. Cham, Switzerland: Springer, 2018, pp. 13–22.

[28] G. Balakrishnan, A. Zhao, M. R. Sabuncu, J. Guttag, and A. V. Dalca, "VoxelMorph: A learning framework for deformable medical image registration," *IEEE Trans. Med. Imag.*, vol. 38, no. 8, pp. 1788–1800, Aug. 2019.

[29] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, 2012, pp. 84–90.

[30] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 740–755.

[31] B. Zhou, H. Zhao, X. Puig, S. Fidler, A. Barriuso, and A. Torralba, "Scene parsing through ADE20K dataset," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 633–641.

[32] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, *arXiv:1502.03167*.

[33] B. Xu, N. Wang, T. Chen, and M. Li, "Empirical evaluation of rectified activations in convolutional network," 2015, *arXiv:1505.00853*.

[34] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[35] H. Sokooti, B. De Vos, F. Berendsen, B. P. Lelieveldt, I. Išgum, and M. Staring, "Nonrigid image registration using multi-scale 3D convolutional neural networks," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2017, pp. 232–239.

[36] A. Van Etten, D. Lindenbaum, and T. M. Bacastow, "SpaceNet: A remote sensing dataset and challenge series," 2018, *arXiv:1807.01232*.

[37] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[38] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.

[39] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[40] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.

[41] K. He, X. Zhang, S. Ren, and J. Sun, "Identity mappings in deep residual networks," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 630–645.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.

[42] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1492–1500.

[43] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[44] C.-Y. Wang, H.-Y. Mark Liao, Y.-H. Wu, P.-Y. Chen, J.-W. Hsieh, and I.-H. Yeh, "CSPNet: A new backbone that can enhance learning capability of CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 390–391.

[45] Y. Lee, J.-W. Hwang, S. Lee, Y. Bae, and J. Park, "An energy and GPU-computation efficient backbone network for real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2019, pp. 1–9.

[46] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[47] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.

[48] Y. Ye, L. Bruzzone, J. Shan, F. Bovolo, and Q. Zhu, "Fast and robust matching for multimodal remote sensing image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9059–9070, Nov. 2019.

[49] J. Li, Q. Hu, and M. Ai, "RIFT: Multi-modal image matching based on radiation-variation insensitive feature transform," *IEEE Trans. Image Process.*, vol. 29, pp. 3296–3310, 2020.

[50] M. Dusmanu et al., "D2-Net: A trainable CNN for joint description and detection of local features," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8092–8101.



Liangzhi Li received the B.E. degree in surveying and mapping engineering from Shandong Jiaotong University, Jinan, China, in 2017, and the master's degree from the School of Geology Engineering and Geomatics, Chang'an University, Xi'an, China, where he is currently pursuing the Ph.D. degree.

His research interests include remote sensing image matching, classification, and segmentation. More details about his work can be found at https://www.researchgate.net/profile/Liangzhi_Li5.



Ling Han received the B.S. and M.S. degrees in remote sensing from Wuhan University, Wuhan, China, in 1991 and 1994, respectively, and the Ph.D. degree in remote sensing from Northwest University, Xi'an, China, in 2005.

Her research interests include the deep learning remote sensing image classification, registration, artificial intelligence, cloud detection in snow-covered images, and landslide interpretation in loess plateau areas.



Mingtao Ding received the B.S., M.S., and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, China, in 2004, 2007, and 2010, respectively.

He was engaged in post-doctoral research at Duke University, Durham, NC, USA, from December 2010 to December 2012, and was invited by The Chinese University of Hong Kong, Hong Kong, to work as a Research Assistant for half a year in 2013. He joined Chang'an University in July 2013, where is currently an Associate Professor with the Department of Surveying and Mapping, School of Geology Engineering and Geomatics.



Hongye Cao was born in Henan, China, in 1989. He received the M.S. degree in surveying and mapping from Southwest Jiaotong University, Chengdu, China, in 2017. He is currently pursuing the Ph.D. degree with Chang'an University, Xian, China.

His research interests include water color remote sensing, mainly including the development of estimation models for water quality parameters, lake cyanobacteria bloom and ocean green tide monitoring via remote sensing, and polluted waters monitoring via remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:00:23 UTC from IEEE Xplore. Restrictions apply.